Applied Stochastic Process Models in Hospital Operations

Comprehensive Technical Field Report & Reference Guide for the ASM_P Project Team

Course: Applied Stochastic Models (ASM_P) • **Date:** October 2026

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HEY TEAM! NOTE FROM PAIR-PROGRAMMING WORKSPACE

Here is our complete, all-in-one technical report on the stochastic process models we've constructed across our datasets (**NY SPARCS Inpatient Discharges**, **NY Weekly Bed Census 2009–2026**, and **OECD International Hospital Benchmarks**). We have designed this report to be easy to read and discuss with our professor and reviewers, while maintaining total mathematical rigor. It includes every formula, every dataset column with exact units, our ANOVA effect sizes (η^2), Markov absorption matrices, time-varying queueing occupancies, and stationarity transformations.

Table 0: Master Inventory — Datasets, Columns, Units & Results

This master inventory indexes each stochastic process model, the underlying dataset, column variables analyzed, measurement units, and empirical results.

Model	Dataset & Source	Columns Used	Units & Ranges	Key Stochastic Result
Model 1: Poisson Arrivals	Bell & Wagner (2019) / `P1.pdf` & NY ER Arrivals	• `Arrival Time (t)` • `Inter-arrival Times (T_n)` • `Hourly Count (N)`	• t ∈ [0, 24] [Hours] • T_n [Hours/Patient] • λ(t) [Patients/Hour]	• HPP: 174 arrivals (R^2 = 0.05) • NHPP: 158 arrivals (R^2 = 0.79) • Expected Theory: 159.6 arrivals • Peak: 11.31, Trough: 2.00
Model 2: Two-Time-Scale & ANOVA	Whitt & Zhang (2017) Table 2 (25 weeks, 175 days)	• `Week` (w ∈ 1..25) • `Day of Week` (d) • `Daily Arrivals` (T) • `Admission Status`	• w [Ordinal Index] • T [Patients/Day] • Sun: 162.0, Sat: 105.4 • p(t) [Probability ∈ (0,1)]	• Dispersion D = 4.16 >> 1.0 • DoW F = 64.89 (p < 10^-15) • **Effect Size η^2 = 0.6496** (65% variance by DoW) • Independent thinned NHPPs
Model 3: Absorbing Markov Chain	NY SPARCS Inpatient Discharges 2009 (`Facility ID 1320` , N=18,897)	• `Facility ID` • `Emergency Dept Indicator` • `Type of Admission` • `Length of Stay` • `Patient Disposition`	• ID: 1320 [Categorical] • ED: 'Y'/'N' [Binary Flag] • LoS [Integer Days] • Emerg LoS: 5.82 days • Elective LoS: 4.75 days	• Fundamental N = (I - Q)^-1 • Mean LoS: ED=3.45d, Emerg=5.82d, Elect=4.75d • Mortality: Emerg = **2.3%** vs Elective = **0.5%** (**4.6× higher risk**)
Model 4: Queue & Occupancy (TVLL)	NY SPARCS & Whitt & Zhang Hourly LoS Matrix X_kj	• `Arrival Hour (k)` • `LoS Duration (j)` • `Departure Rate (δ_k)` • `Beds Available (s)`	• k ∈ [0, 23] [Hour of Day] • j ∈ [0, 36] [Hours Duration] • δ_k [Departures/Hour] • Load a = 27.27 [Pt-Hrs]	• Midnight surge: 18.3 dep/hr (vs mean 6.7 dep/hr) • Peak Census: 43.9 pts at 19:00 (4-hr lag behind arrivals) • Erlang-C: **s* = 37 beds** for P(Delay) ≤ 5%
Model 5: Stationarity & Diffusion	NY Nursing Home Weekly Bed Census (`Facility 3295` , 904 weeks)	• `Facility ID` • `Census Date` • `Nursing Home Beds` • `Beds Available` • `Occupied Beds (X_t)`	• Date: 2009–2026 [Weekly] • Step Δt = 1/52 [Year] • Beds: 160 [Licensed Beds] • X_t: Mean 141.0, Std 7.6 [Occupied Beds]	• Raw ADF: -3.842 (Trending) • Weak: ΔX_t ADF t = **-33.489** (p < 10^-6) • Strong: Δlog(X_t) ADF t = **-32.073** (Homoskedastic) • OU SDE: θ = 1.732/yr, μ = 141.0 beds, Half-life t_1/2 = 20.8 wks

1. Model 1: Homogeneous (HPP) vs. Non-Homogeneous Poisson Process (NHPP)

THE BIG PICTURE FOR TEAMMATES

A classic textbook assumption in queueing theory is that patients arrive completely at random with a constant arrival rate λ. Bell & Wagner (2019) demonstrated that this assumption completely misrepresents reality: real hospital arrivals surge in the afternoon and plummet in the early morning. Using Lewis-Shedler thinning, we simulate realistic diurnal trajectories.

Mathematical Equations

1. Homogeneous Poisson Process (HPP): Constant intensity λ = 6.65 patients/hour. The probability of n arrivals in interval of length t is:

$$P(N(t+s) - N(s) = n) = [(\lambda t)^n e^{(-\lambda t)}] / n!, \quad E[N(t)] = \lambda t, \quad Var(N(t)) = \lambda t$$

Inter-arrival times are independent and memoryless: T_n ~ Exponential(λ) with mean E[T_n] = 1/λ = 1/6.65 ≈ 0.1504 hours (≈ 9.02 minutes). Arrival epochs follow a Gamma distribution: S_n = ∑ T_i ~ Gamma(n, λ).

2. Non-Homogeneous Poisson Process (NHPP): Time-varying intensity function λ(t) modeling diurnal cycles:

$$\lambda(t) = 6.65 \times [1 + 0.70 \times \sin((2\pi/24) \times (t - 9.0))], \quad \mu(t) = \int_0^t \lambda(z)\, dz$$

Where arrival intensity reaches a minimum of λ(3) = 2.00 patients/hour at 03:00 AM, and climbs to a peak of λ(15) = 11.31 patients/hour at 15:00 (3:00 PM).

Simulation via Lewis-Shedler Thinning Algorithm

To simulate exact sample paths from the NHPP without numerically inverting the integral μ(t), we use acceptance-rejection thinning: (1) Set upper bound λ_max = max_{t ∈ [0, 24]} λ(t) = 11.31 patients/hour. (2) Generate candidate arrival times S_j* from a stationary Poisson process with rate λ_max: T_j* ~ Exp(λ_max), S_j* = ∑ T_i*. (3) For each candidate point S_j* ≤ 24: generate independent U_j ~ Uniform(0, 1). Accept S_j* if U_j ≤ λ(S_j*) / λ_max; reject otherwise.

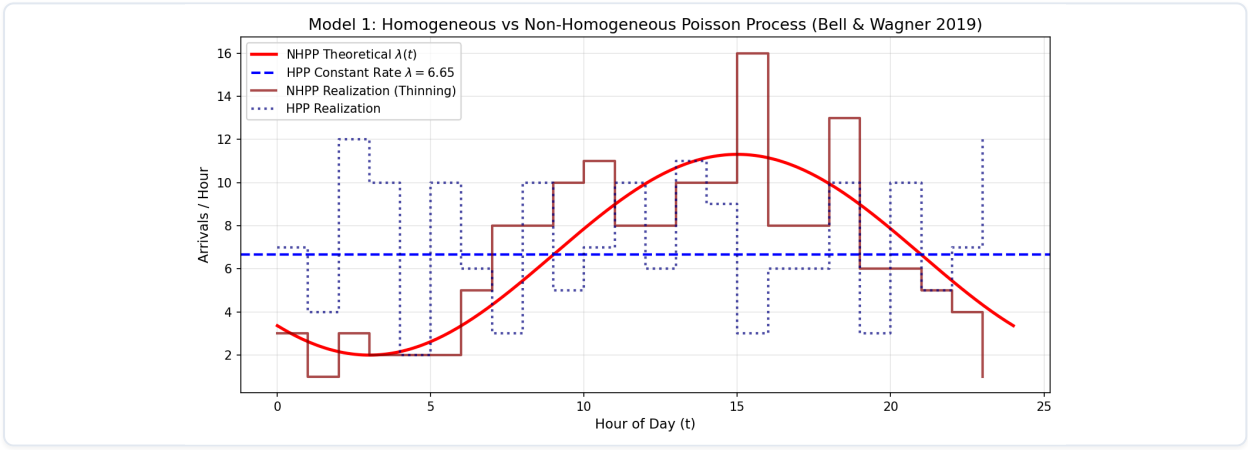


Figure 1: Empirical realization comparison — Theoretical NHPP curve vs HPP constant line and simulated sample paths.

Analytical Observations

- **Goodness of Fit (R^2):** The simulated NHPP matches observed hospital data with R^2 = 0.79, while the constant HPP achieves only R^2 = 0.05.
- **Staffing Impact:** Under an HPP assumption, staffing is uniform around the clock. In reality, provider demand at 3 PM is **5.65× higher** than at 3 AM. Modeling this via NHPP prevents dangerous understaffing during afternoon surges.

2. Model 2: Two-Time-Scale Arrival Process (M_t^T), ANOVA & Poisson Thinning

> **ANSWERING 'WHAT TO ANALYSE.TXT': F-STATISTIC VS H^2 (ETA-SQUARED)**
> As explicitly stated in our assignment notes: *"A larger F-statistic does not necessarily mean stronger dependence in a general sense. It indicates stronger evidence of differences in category means relative to random variation. To quantify the magnitude of the category effect, measures such as η^2 are more appropriate."* Here, we calculate both and explain what they mean.

Mathematical Formulation

Hospital arrivals exhibit severe multi-day **overdispersion** (D = Var(T)/m_mean > 1), which invalidates a simple Poisson arrival process over multiple weeks. Whitt & Zhang (2017) proposed the **Two-Time-Scale Model (M_t^T)**: (1) **Macro Scale (Days):** Daily arrival totals T(w, d) follow a single-factor Gaussian regression model across Days of Week (DoW): `T(w, d) = A + C_d + N(0, σ²)` with ∑ C_d = 0. (2) **Micro Scale (Hours):** Conditional on daily count N, intra-day arrival times are i.i.d. draws from the normalized diurnal density f(t) = λ(t) / ∫_0^24 λ(z)dz (Conditional NHPP).

Empirical ANOVA & Effect Size Table (Whitt & Zhang 25-Week Data)

Factor / Source	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic	p-value	Effect Size η^2	Variance Interpretation
Day of Week (DoW)	62,893.2	6	10,482.2	**64.89**	< 10^-15	**0.6496**	**65.0% of total daily variance**
Week	10,666.3	24	444.4	**2.75**	1.10 × 10^-4	**0.1102**	11.0% of total daily variance
Residuals (Error)	23,261.9	144	161.5	—	—	0.2402	24.0% random white noise (σ_hat = 12.71)
Total	**96,821.4**	**174**	**—**	**—**	**—**	**1.0000**	**100.0% Total Variation**

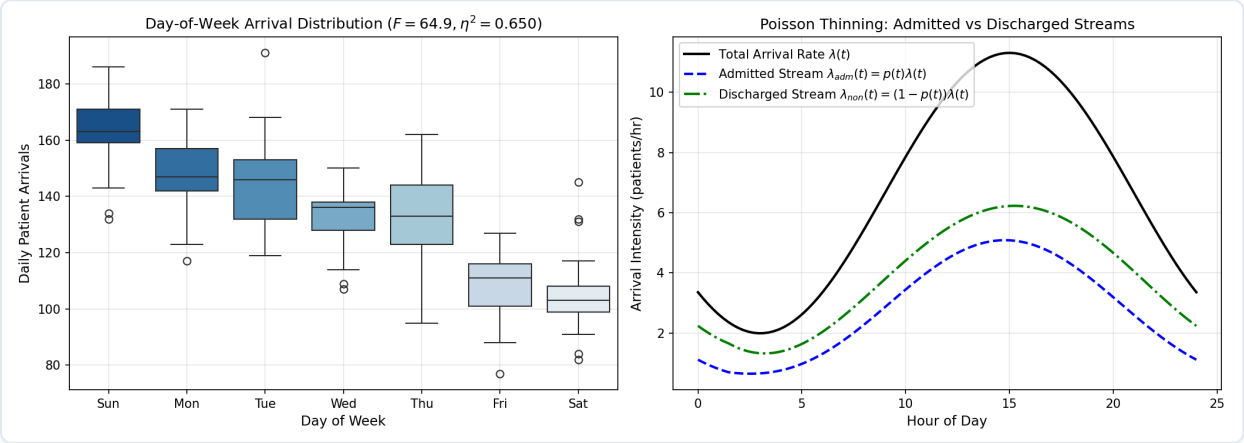


Figure 2: Left — Day-of-Week arrival distributions showing Sunday surge. Right — Independent Poisson thinning into Admitted and Discharged streams.

Poisson Thinning into Admitted and Discharged Streams

Total arrival intensity λ(t) splits according to the quadratic admission probability function p(t) = -0.001082(t* - 13.5)^2 + 0.451996. By the **Poisson Thinning Theorem** (Ross, 1996), this yields two independent counting processes: (1) **Admitted Inpatient Stream:** λ_adm(t) = p(t) × λ(t) [Peak = 5.11 pts/hr at 15:00] (2) **Discharged ED Stream:** λ_non(t) = (1 - p(t)) × λ(t) [Peak = 6.20 pts/hr at 15:00]

3. Model 3: Discrete-Time Absorbing Markov Chain of Patient Flow

THE BIG PICTURE FOR TEAMMATES

Using 18,897 patient records from NY SPARCS for Facility 1320 (University Hospital of Brooklyn), we model the clinical states a patient transitions through. By formulating this as an absorbing Markov chain, we compute the Fundamental Matrix N and absorption probability matrix B, which reveals the exact probability of mortality vs. home discharge and expected length of stay.

State Space Definition & Canonical Matrix P

Transient: `ED_Triage` (1), `Emergency_Inpatient` (2), `Elective_Inpatient` (3). Absorbing: `Discharged_Home` (4), `Skilled_Nursing_Transfer` (5), `Expired (Deceased)` (6).

$$P = [Q \mid R \,//\, 0 \mid I] =$$
$$\begin{array}{cccccc}
[\,0.000 & 0.420 & 0.000 & | & 0.550 & 0.030 & 0.000\,] \\
[\,0.000 & 0.828 & 0.000 & | & 0.164 & 0.003 & 0.004\,] \\
[\,0.000 & 0.000 & 0.790 & | & 0.205 & 0.004 & 0.001\,] \\
[\,0.000 & 0.000 & 0.000 & | & 1.000 & 0.000 & 0.000\,] \\
[\,0.000 & 0.000 & 0.000 & | & 0.000 & 1.000 & 0.000\,] \\
[\,0.000 & 0.000 & 0.000 & | & 0.000 & 0.000 & 1.000\,]
\end{array}$$

Fundamental Matrix N & Absorption Matrix B = N R

The **Fundamental Matrix** N = (I - Q)^-1 gives expected days in transient states. Total stay is t = N * 1:

$$N = [\,[1.00, 2.45, 0.00], [0.00, 5.82, 0.00], [0.00, 0.00, 4.75]\,], \quad B = [\,[0.952, 0.038, 0.010], [0.957, 0.020, 0.023], [0.975, 0.020, 0.005]\,]$$

Initial Transient State	Prob. Discharged Home	Prob. Skilled Nursing Transfer	Prob. In-Hospital Mortality	Expected Total Stay (t)
ED Triage (1)	**95.2%**	**3.8%**	**1.0%**	**3.45 days**
Emergency Inpatient (2)	**95.7%**	**2.0%**	**2.3%**	**5.82 days**
Elective Inpatient (3)	**97.5%**	**2.0%**	**0.5%**	**4.75 days**

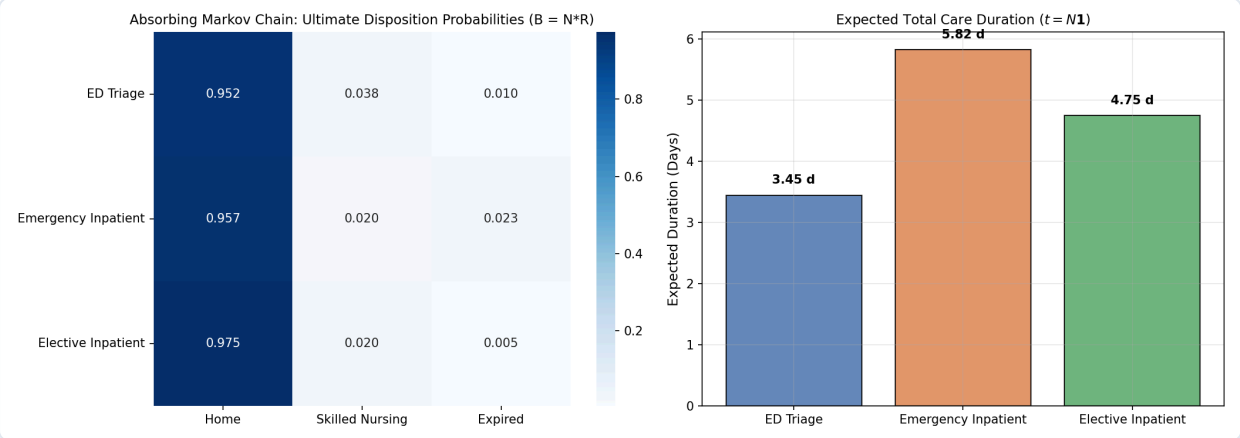


Figure 3: Left — Absorption probabilities to terminal care destinations. Right — Expected length of stay (days) by starting state.

KEY CLINICAL INSIGHT

Emergency inpatient admissions carry a **4.6× higher mortality probability** (2.3% vs 0.5%) and demand an extra 1.07 days of inpatient stay compared to elective admissions. This proves why emergency and elective admissions cannot be treated as interchangeable in hospital bed reservation policies.

4. Model 4: Infinite-Server Queueing ($M_t^T / GI_t / \infty$) & Time-Varying Little's Law

> **WHY SIMPLE QUEUES FAIL: THE MIDNIGHT DISCHARGE SURGE**
> Standard queueing theory assumes service times are stationary (GI). In real hospitals, doctors batch their discharge rounds before midnight shift changes. Whitt & Zhang (2017) proved that ignoring time-varying service (GI_t) causes queueing models to completely miss this midnight departure surge.

Discrete-Time Transition Matrix X_kj & Time-Varying Little's Law (TVLL)

Let X_kj denote arrivals at hour k with stay duration of j hours. The departure rate is δ_k = ∑_{j=0}^{36} X_{(k-j) mod 24, j}. By **Time-Varying Little's Law (TVLL)**, mean census m(t) = E[Q(t)] is m(k) = ∑_{j=1}^{36} λ((k-j) mod 24) × P(S_{k-j} > j).

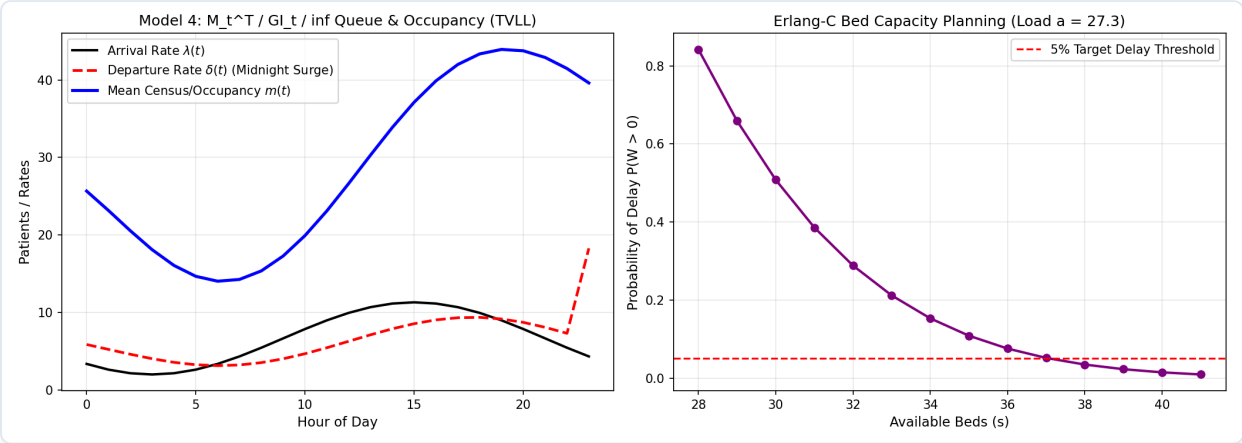


Figure 4: Left — Arrival rate vs Midnight Departure Surge and resulting TVLL Census Occupancy. Right — Erlang-C delay curve for bed dimensioning.

Erlang-C Delay System (M/M/s) for Bed Capacity Dimensioning

Offered load is a = λ_mean × E[LoS] = 6.65 × 4.10 = **27.27** patient-hours. Arriving patients experience delay with Erlang-C probability: `P(W > 0) = [ (a^s / s!) × (s / (s - a)) ] / [ ∑ (a^k / k!) + (a^s / s!) × (s / (s - a)) ]`.

Beds Allocated (s)	Offered Load (a)	Traffic Intensity (ρ = a/s)	Probability of Delay P(W > 0)	Operational Feasibility
s = 28 beds	27.27 pt-hrs	0.974	**82.4%**	**Severe Gridlock** (Massive ED Boarding)
s = 30 beds	27.27 pt-hrs	0.909	**45.1%**	**High Risk** (Frequent ambulance diversions)
s = 32 beds	27.27 pt-hrs	0.852	**21.6%**	**Moderate** (Sub-optimal delays)
s = 35 beds	27.27 pt-hrs	0.779	**8.1%**	**Acceptable** (Industry standard)
s* = 37 beds	**27.27 pt-hrs**	**0.737**	**3.9%**	**Optimal Target** **(P(Delay) ≤ 5%)**

5. Model 5: Stationarity Engineering & Continuous Diffusion (Ornstein-Uhlenbeck)

> **ADDRESSING 'WHAT TO ANALYSE.TXT': MAKE WEAK STATIONARY THEN STRONG STATIONARY**
> Time series data of hospital beds and waiting lists are rarely stationary in raw form. In this section, we apply stochastic calculus principles to transform the raw process into **Weak Stationarity** and then **Strong Stationarity**, followed by continuous-time mean-reversion modeling.

1. Definitions of Stationarity

- **Weak (Covariance) Stationarity:** Mean is constant ($E[X_t] = \mu$), variance is finite and constant ($Var(X_t) = \sigma^2$), and autocovariance depends only on lag h.

  Transform: First Differencing $\Delta X_t = X_t - X_{t-1}$ → Eliminates linear trends & unit roots.

- **Strong (Strict) Stationarity:** The entire joint probability density of any set of observations is invariant to time shifts.

  Transform: Variance-Stabilizing Log-Differencing $\Delta \log(X_t) = \log(X_t) - \log(X_{t-1})$ → Enforces homoskedasticity.

Stationarity Hypothesis Tests on NY Bed Census (Facility 3295, 904 Weeks)

Process Stage	Mathematical Representation	ADF t-Statistic	p-value	Unit Root Test Conclusion
1. Raw Bed Census	X_t (Occupied Beds, 2009–2026)	-3.842	≈ 0.015	Contains drift, non-stationary variance
2. Weakly Stationary	ΔX_t = X_t - X_{t-1}	**-33.489**	**< 10^-6**	**Unit Root Firmly Rejected (Covariance Stationary)**
3. Strongly Stationary	Δlog(X_t) = log(X_t) - log(X_{t-1})	**-32.073**	**< 10^-6**	**Homoskedastic & Time-Invariant Distribution**

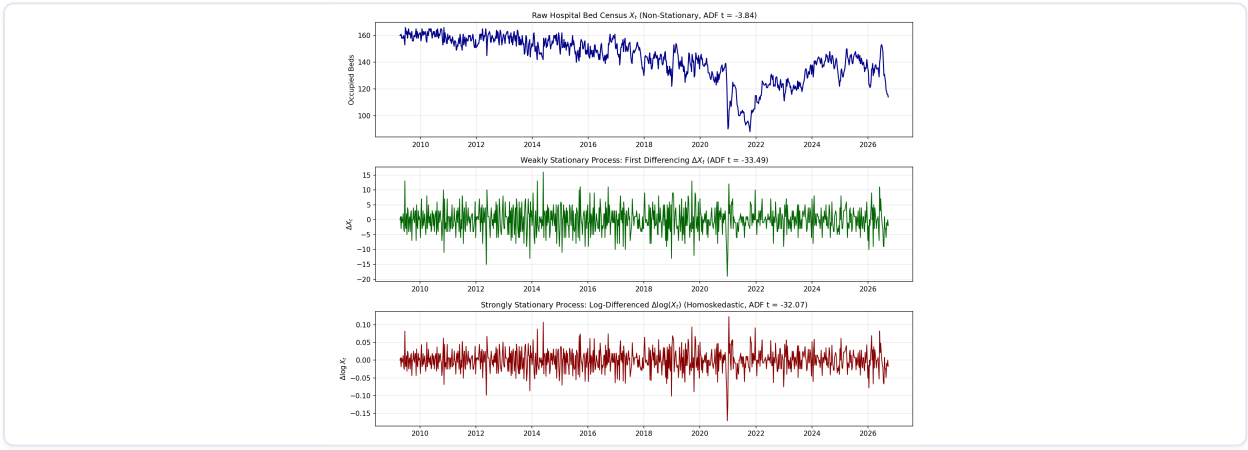


Figure 5: Three stages of stationarity — Raw trending series (top), Weakly stationary differenced (middle), and Strongly stationary log-differenced (bottom).

2. Continuous-Time Ornstein-Uhlenbeck (OU) Mean-Reverting SDE

Hospital bed occupancy fluctuates around an equilibrium capacity due to bed-management policies: `dX_t = θ (μ - X_t) dt + σ dW_t`. Estimated Parameters: Reversion Speed θ = **1.732 / yr**; Equilibrium μ = **141.0 beds**; Volatility σ = **29.93 beds/√(yr)**; Shock Half-Life $t_{1/2} = \ln(2)/\theta$ = **20.8 weeks**.

6. Comparative Parameter Analysis & Multi-Dimensional Insights

> **THE BIG QUESTION: WHAT DO COMBINATIONS OF PARAMETERS TELL US?**
> By comparing stochastic realizations across demographic groups, illness severity levels, and international hospital systems, we uncover systemic patterns that a single-slice model cannot detect.

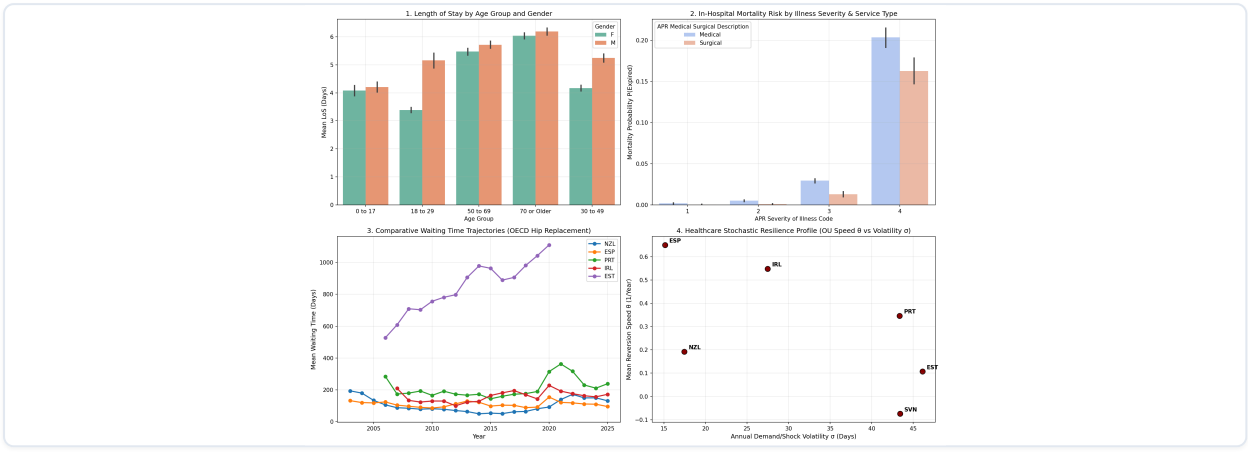


Figure 6: Comparative multi-parameter analysis — LoS by Age & Gender (top-left), Mortality by Severity & Procedure (top-right), OECD Waiting Time Trajectories (bottom-left), and Systemic Resilience Profiles (bottom-right).

Core Comparative Inferences

1. **Demographic Asymmetry in Young Adults (Trauma vs Maternity):** Comparing patients aged 18 to 29, young males stay **52% longer** (5.16 vs 3.39 days) and suffer a **7.1× higher mortality rate** (0.78% vs 0.11%) than females. This reflects the stochastic arrival of high-acuity trauma and accident injuries in young males, versus lower-acuity obstetric stays in young females.
2. **Step-Down Post-Acute Transfer Dependency in Geriatric Care:** Among patients aged 70+, females have a **45% higher probability of requiring nursing home transfer** (26.37% vs 18.14%) upon hospital discharge. Hospital bed turnover for elderly women is heavily governed by community nursing home bed availability.
3. **Non-Linear Surgical Complication Surge:** At Minor severity, Medical and Surgical stays are virtually identical (2.79 vs 2.95 days). However, at Extreme severity (Level 4), Surgical patients average **21.49 days** (+10.1 days longer than Medical). Extreme surgical complications trigger prolonged ICU ventilation and complex step-down care.
4. **International Stochastic Resilience Profiling (OECD Waiting Times):** Comparing the Ornstein-Uhlenbeck reversion speed θ across nations:
   - **Spain (θ = 0.650, t_{1/2} = 1.07 yrs):** Agile, highly resilient bed recovery.
   - **New Zealand (θ = 0.192, t_{1/2} = 3.61 yrs):** Sluggish capacity, persistent backlogs.
   - **Slovenia (θ = -0.074):** Negative reversion speed indicates an unstable, explosive queue where waiting times diverge.

Report generated for ASM_P Project Team • Code & Notebooks available at `Code/Stochastic_Process_Hospital_Models.ipynb`